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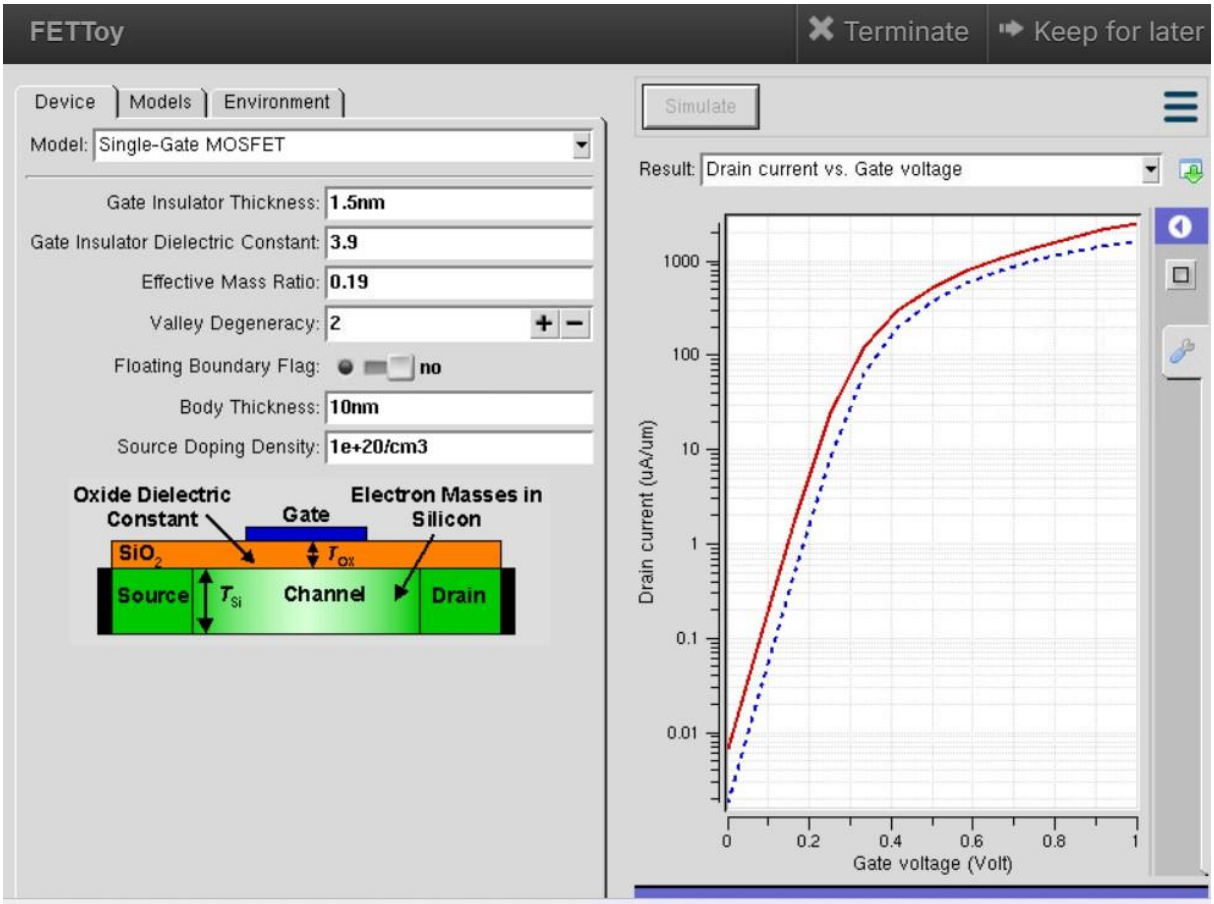
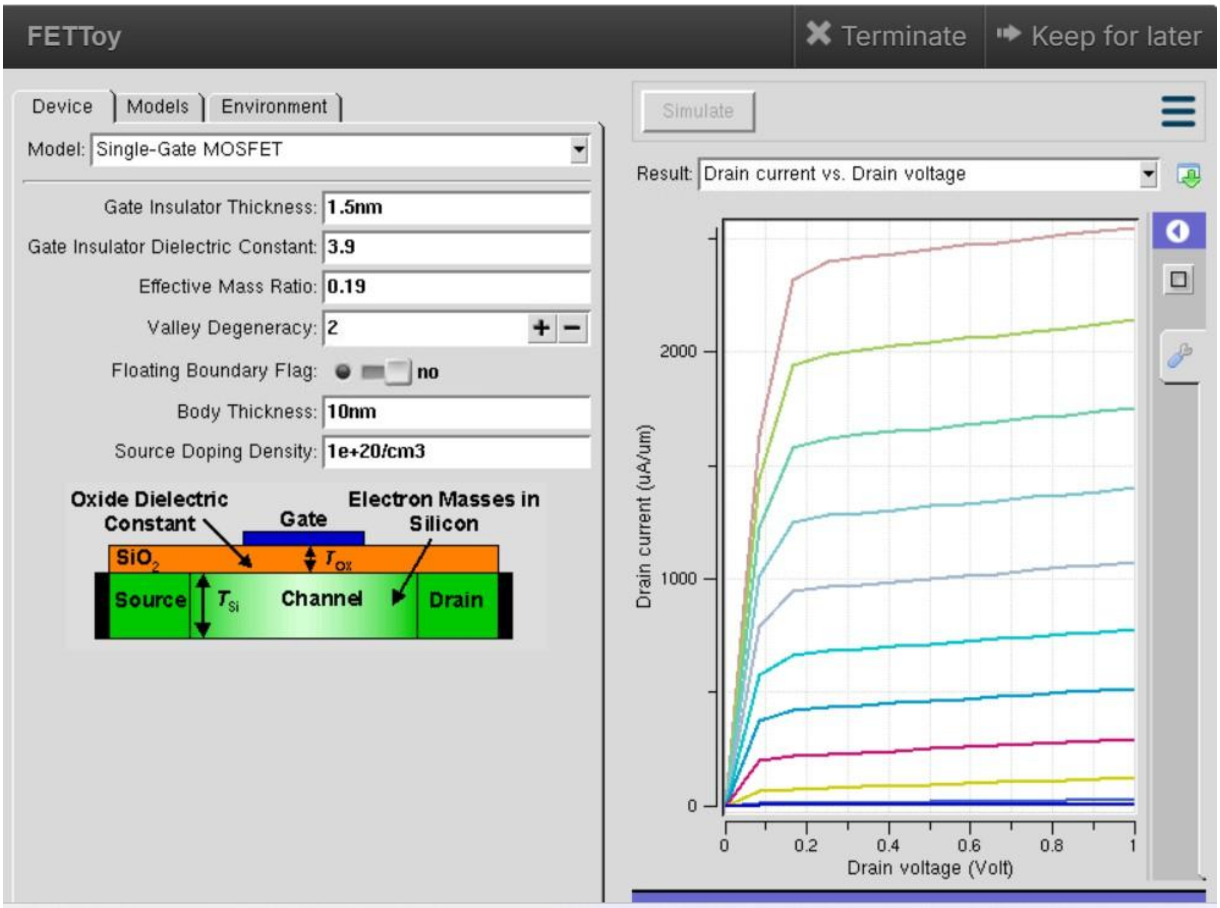
NANO ELECTRONICS

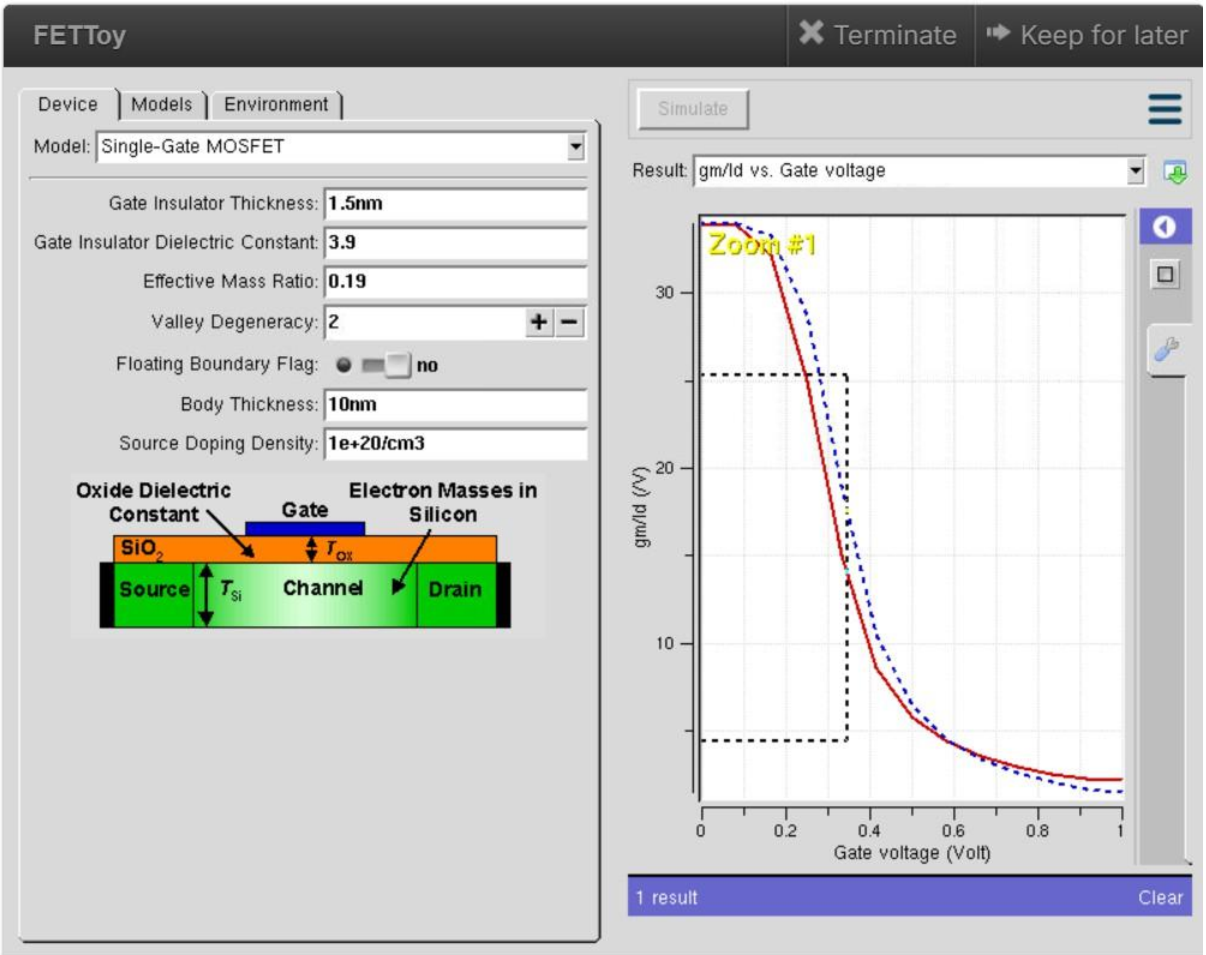
Exp. 6. MOSFET Scaling Analysis and Subthreshold Swing Calculation using NanoHub



This figure shows the FETToy simulation of a Single-Gate MOSFET with a 1.5 nm gate oxide thickness**,** dielectric constant of 3.9**,** and source doping of 1×10^{20} cm^{-3}. The left side illustrates the MOSFET structure, while the graph on the right shows the Drain Current (Id) versus Gate Voltage (Vg**)**. The drain current increases with gate voltage, indicating effective channel formation and improved carrier transport. This simulation is useful for analysing MOSFET performance and optimising CMOS and low-power semiconductor devices**.**



This figure shows the FETToy simulation of a Single-Gate MOSFET, illustrating the Drain Current (Id) versus Drain Voltage (Vd) for different Gate Voltage (Vg) values. The output curves show that the drain current increases with drain voltage and reaches saturation. Higher gate voltage improves channel conductivity, resulting in greater drain current. This simulation is useful for evaluating MOSFET performance and optimising CMOS and low-power semiconductor devices.



This image illustrates the FETToy simulation results of a Single-Gate MOSFET, depicting the relationship between Transconductance Efficiency (gm/Id) and Gate Voltage (Vg). The plotted curves compare two simulation characteristics, showing that the gm/Id ratio gradually decreases as the gate voltage increases, indicating changes in the device's operating efficiency across different bias regions. Higher gm/Id values at lower gate voltages signify improved transconductance efficiency, which is advantageous for low-power and low-voltage circuit design. This simulation is commonly employed to analyze and optimize MOSFET characteristics for analog circuit design, CMOS applications, integrated circuit development, and energy-efficient nanoelectronic systems.